

MA540 Project 2

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Problem 1

Statement

We analyze the steady state heat model for a copper rod, which follows the boundary value problem

$$\frac{d^2 T_s}{dx^2} = \frac{2(a+b)}{ab} \frac{h}{k} [T_s(x) - T_{amb}]$$
$$\frac{dT_s}{dx}(0) = \frac{\Phi}{k} \quad , \quad \frac{dT_s}{dx}(L) = \frac{h}{k} [T_{amb} - T_s(L)]$$

We can use the known thermal conductivity $k = 4.01 \frac{W}{cm \cdot ^\circ C}$, and we take $u_{amb} = 22.28$. The cross-sectional dimensions of the rod are $a = b = 0.95$ cm with length $L = 70$ cm. We will use the sensitivity relations $\frac{\partial f}{\partial \Phi}$ and $\frac{\partial f}{\partial h}$ found through complex-step approximation on Project 1, Problem 3, along with a least-squares minimization to find a best fit model. We will compare this with the data in Table 3.3:

x (cm)	10	14	18	22	26	30	34	38
Temp ($^\circ C$)	66.04	60.04	54.81	50.42	46.74	43.66	40.76	38.49
x (cm)	42	46	50	54	58	62	66	
Temp ($^\circ C$)	36.42	34.77	33.18	32.36	31.56	30.91	30.56	

Finding residuals between this data and the modeled data will allow us to calculate an estimate of the variance by $\sigma^2 = \frac{R^T R}{n-p}$, where $n = 15$ (number of data points), $p = 2$ (number of parameters of interest), and R is the vector of residuals. We will use this to find the covariance matrix by $V = \sigma^2 (X^T X)^{-1}$, where X is the sensitivity matrix. We will then use these same values to create a Q-Q plot.

Solution

With $\sigma^2 = 5.29 * 10^{-2}$, we construct the covariance matrix:

$$V = \begin{bmatrix} V(\frac{\partial f}{\partial \Phi}) & Cov(\frac{\partial f}{\partial \Phi}, \frac{\partial f}{\partial h}) \\ Cov(\frac{\partial f}{\partial h}, \frac{\partial f}{\partial \Phi}) & V(\frac{\partial f}{\partial h}) \end{bmatrix} = \begin{bmatrix} 8.96 * 10^{-3} & -1.21 * 10^{-6} \\ -1.21 * 10^{-6} & 1.77 * 10^{-10} \end{bmatrix}$$

The covariance of Φ and h is low, which implies that these parameters aren't correlated. Next, we plot the residuals, comparing our best fit model with the given data (Figure 1). Residuals appear both positive and negative, with a noticeable upside-down "V" pattern. This may show a slight bias

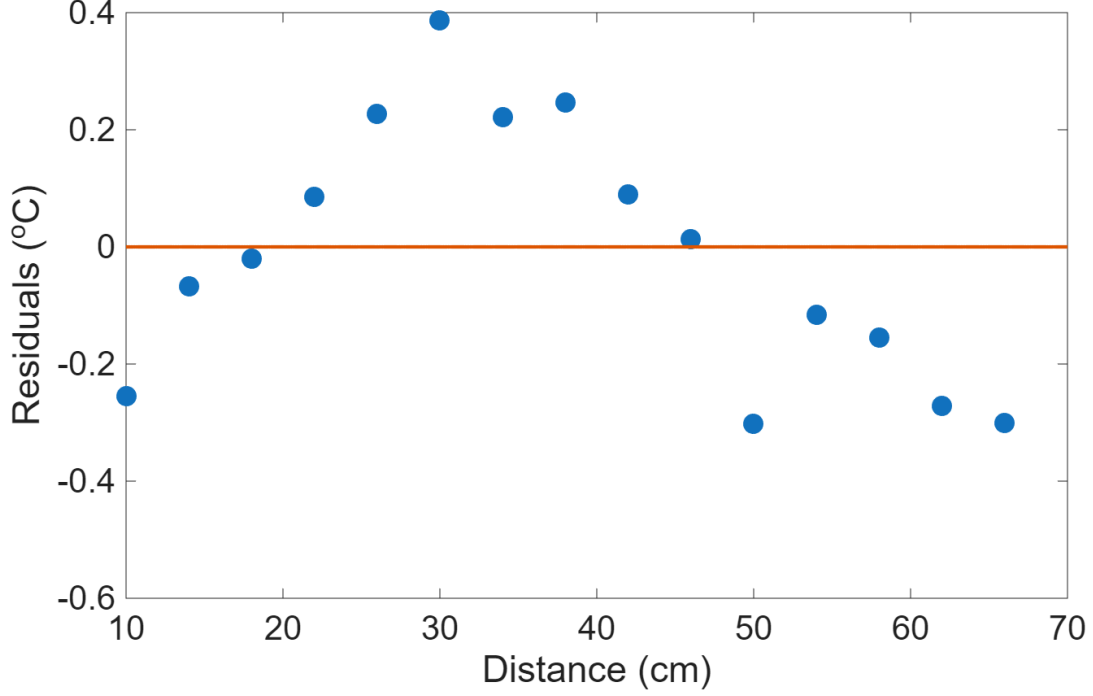


Figure 1: T Residuals

in our model, but considering the small sample size, it is not unreasonable to conclude that the error is iid. Finally, we create a Q-Q plot, comparing the distribution of our residuals with the quantiles of a normal distribution (Figure 2). The Q-Q plot closely follows a diagonal, which is the line where the quantiles are equivalent. This shows that our residuals do align with a normal distribution. Hence, we can say that the error is normally distributed.

Problem 2

Statement

We will now analyze the Helmholtz energy

$$\psi(P, \theta) = \alpha_1 P^2 + \alpha_{11} P^4 + \alpha_{111} P^6,$$

where P is the polarization, with nominal values $\alpha_1 = -382.7$, $\alpha_{11} = 760.3$ and $\alpha_{111} = 155.5$. (a) First, we will compute the OLS estimate for the observation variance σ^2 for three separate sets of observations, following $P_i = (i-1)\Delta P$, $\Delta P = \frac{0.8}{n-1}$, $i = 1, \dots, n$ for $n = 81, 161$ and 801 . We will add a normally-distributed error, such that

$$Y_i = \psi(P_i, \theta) + \varepsilon_i,$$

where $\varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = 12.96$. The design matrix X for the OLS estimate will be a $n \times 3$ matrix, where the columns are P_i^2, P_i^4, P_i^6 , respectively. Then, $\theta = (X^T X)^{-1} X^T Y$, where Y is the matrix of Y_i s. We then use $R = Y - X\theta$ to find residuals, and again use $\sigma^2 = \frac{R^T R}{n-p}$ to calculate our OLS estimate of the variance, with $p = 3$. (b) Next, we will consider synthetic data from the file

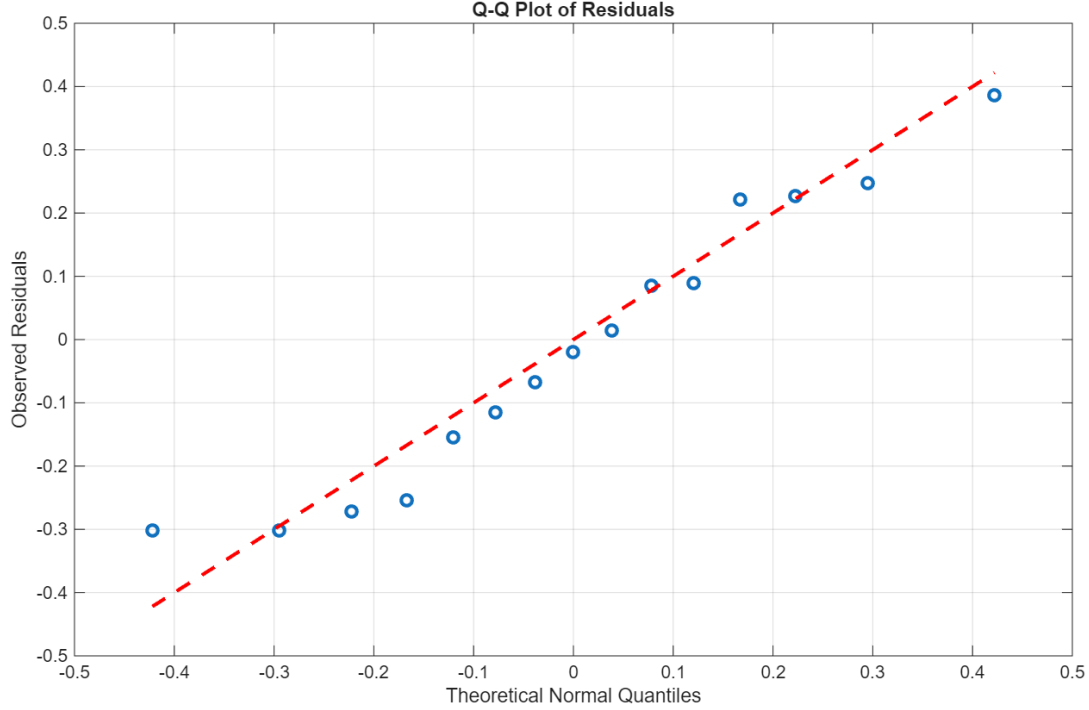


Figure 2: Q-Q Plot of T Residuals

`Helmholtz.txt`, estimating the parameters θ , observation variance σ^2 , and covariance matrix V . The parameters will be optimized using `fminsearch` on a least-squares function, and the variance and covariance matrix found by the same calculations as Problems 1 and 2a. This time, the energy values from the synthetic data will be inputted to form our design matrix. We will then plot the residuals and 2σ intervals, and discuss observations. (c) Finally, we will use

$$\left[\hat{Y}_{x_*} \pm t_{n-p, 1-\alpha/2} \cdot \hat{\sigma} \sqrt{1 + x_*(X^T X)^{-1} x_*^T} \right]$$

to compute 95% prediction intervals for the response, and plot them together with our model and observations. The t-value for this will be found using an inverse Student T-distribution calculator.

Solution

(a) Due to random sampling, the specific estimates vary between each run, but a pattern emerges when comparing to the true value of $\sigma^2 = 12.96$:

n = 81	n = 161	n = 801
1.211	2.496	13.35
1.088	2.258	12.65
1.159	2.819	12.08

As n increases, the estimated σ^2 converges to the true value. This follows, as we can expect to converge to the true value as we increase our sample size. This also indicates that the OLS estimate increases in accuracy at large sample sizes.

(b) Using least squares, we estimate the optimal parameters to be $[\alpha_1, \alpha_{11}, \alpha_{111}] = [-382.7, 760.3, 155.5]$. We note that these are equivalent to the nominal values. We find the observation variance to be

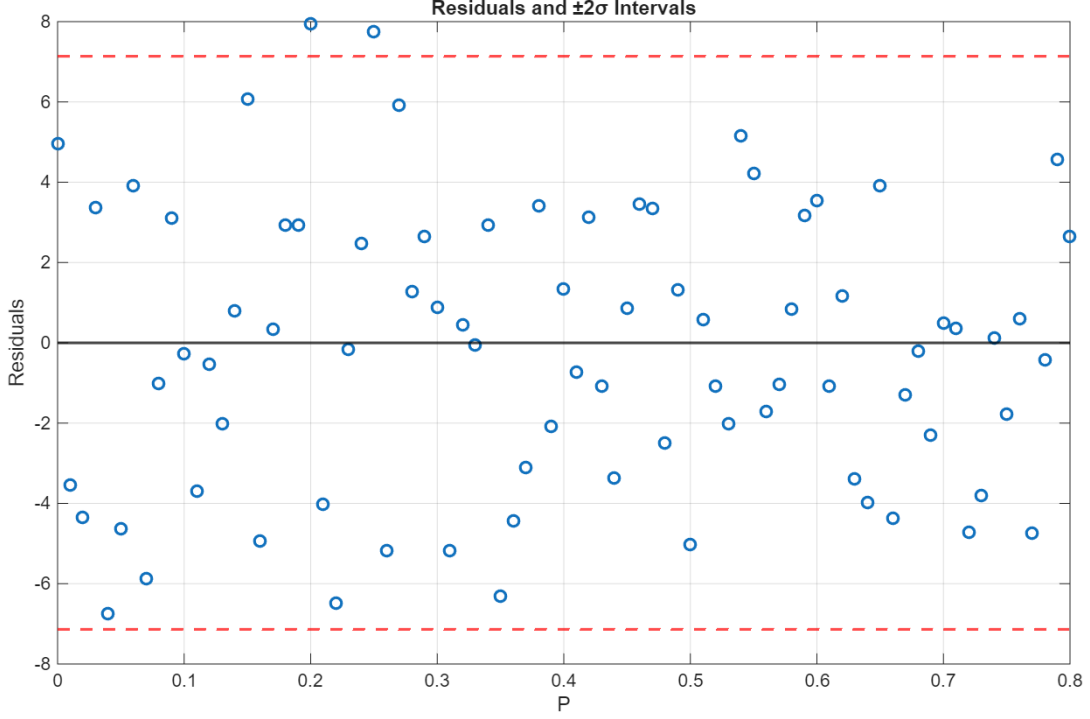


Figure 3: Residuals with 2σ intervals

$\sigma^2 = 12.7367$. And we construct the covariance matrix:

$$V = \begin{bmatrix} V(\frac{\partial\psi}{\partial\alpha_1}) & Cov(\frac{\partial\psi}{\partial\alpha_1}, \frac{\partial\psi}{\partial\alpha_{11}}) & Cov(\frac{\partial\psi}{\partial\alpha_1}, \frac{\partial\psi}{\partial\alpha_{111}}) \\ Cov(\frac{\partial\psi}{\partial\alpha_{11}}, \frac{\partial\psi}{\partial\alpha_1}) & V(\frac{\partial\psi}{\partial\alpha_{11}}) & Cov(\frac{\partial\psi}{\partial\alpha_{11}}, \frac{\partial\psi}{\partial\mu}) \\ Cov(\frac{\partial\psi}{\partial\alpha_{111}}, \frac{\partial\psi}{\partial\alpha_1}) & Cov(\frac{\partial\psi}{\partial\alpha_{111}}, \frac{\partial\psi}{\partial\alpha_{11}}) & V(\frac{\partial\psi}{\partial\alpha_{111}}) \end{bmatrix} = \begin{bmatrix} 1.17 * 10^2 & -5.68 * 10^2 & 6.33 * 10^2 \\ -5.68 * 10^2 & 3.0 * 10^3 & -3.52 * 10^3 \\ 6.33 * 10^2 & -3.52 * 10^3 & 4.27 * 10^3 \end{bmatrix}$$

Notably, the covariance values are extremely large, which shows that these parameters are highly correlated. Global sensitivity methods often assume mutually independent parameters to streamline calculations, but in cases such as this, such an assumption would be inappropriate. Hence, results based on that assumption may be misleading. In Figure 3, we plot the residuals with 2σ intervals. Visually, the vast majority of the data is contained within the intervals, which follows the 95% expectation. As such, the intervals appear to be correct.

(c) Figure 4 shows the plotted model, observations, and 95% prediction intervals. As shown, the model follows the data closely, and the intervals fall appropriately alongside the data. Our model appears to be sufficient for predicting the energy relation.

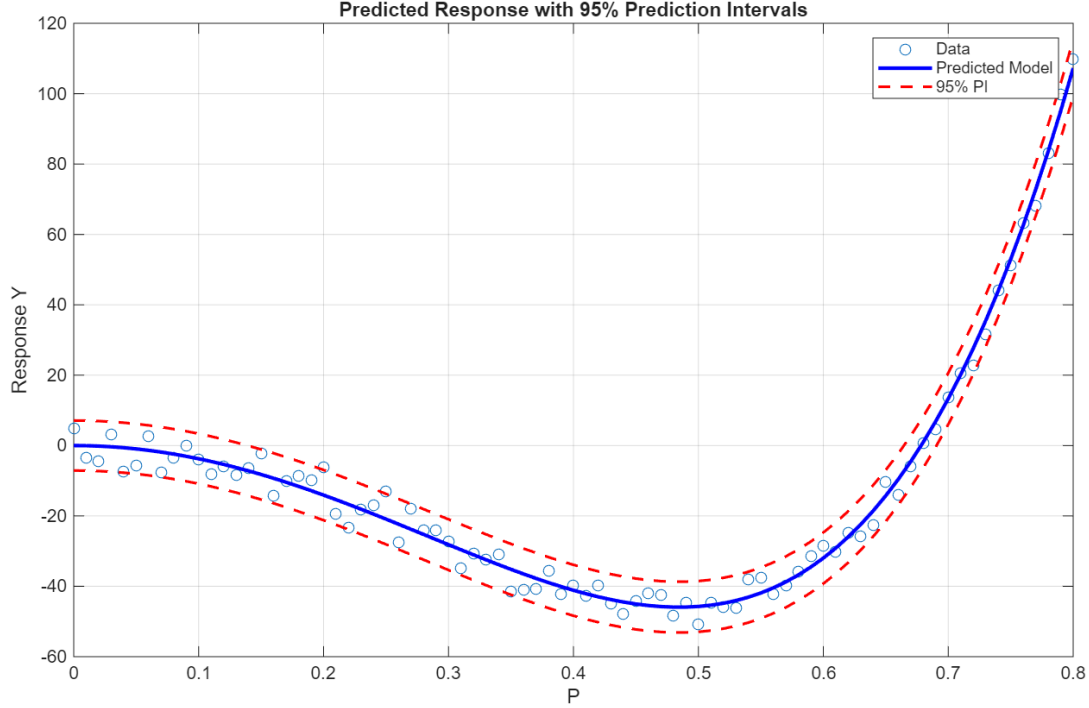


Figure 4: Predicted Response with 95% intervals

Problem 3

Statement

We consider the SIR model

$$\begin{aligned} \frac{dS}{dt} &= \mu(N - S) - \beta IS \quad , \quad S(0) = 900, \\ \frac{dI}{dt} &= \beta IS - (\gamma + \mu)I \quad , \quad I(0) = 100, \\ \frac{dR}{dt} &= \gamma I - \mu R \quad , \quad R(0) = 0, \end{aligned}$$

where the parameters β, γ and μ are each in the interval $[0, 1]$. (a) We will use the data in the file `SIR_integer.txt` to estimate these parameters by least squares, using `fmincon.m` to restrict the solutions to the interval $[0, 1]$. We use `ode15s.m` to approximate the solution. (b) Next, we construct the local sensitivity matrix using the complex-step approximation, estimate the variance σ^2 of the observations, compute the covariance matrix $\mathbf{V}$, and discuss the parameter correlation. These calculations will use the same methods as in previous problems. Here, $n = 151$ and $p = 3$. (c) Next, we will compute 95% confidence intervals, through the same method as in Problem 2c, and compare to intervals of two standard deviations about the parameter estimates. (d) With these parameter estimates and variances, we can plot the distributions for each of the three parameters, using `normpdf.m` to input our estimates. We also plot the residuals, model fit to the data $I(t_j)$, and the trajectories for $S(t), I(t)$ and $R(t)$. (e) Lastly, we consider a different set of data collecting during a flu outbreak at a British boarding school, where $N = 733$, with $S(0) = 730, I(0) = 3$ and $R(0) = 0$. We assume $\mu = 0$ due to the brief duration of the outbreak, and consider the parameters $\theta = [\beta, \gamma]^T$. The data is listed in Table 1. We will estimate the parameters fitting this data, again

Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Confined to Bed	3	8	26	76	225	298	258	233	189	128	68	29	14	4

Table 1: Influenza data reported in [370] from the source [12].

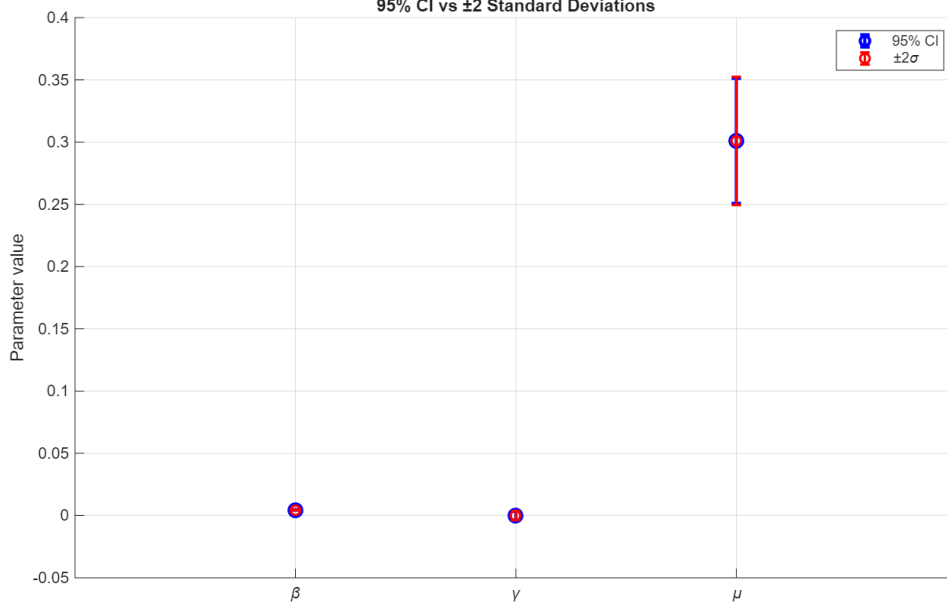


Figure 5: Confidence Intervals vs $\pm 2\sigma$

by least squares, as well as estimating the variance and covariance matrix, again using complex-step approximation. Here, $n = 14$ and $p = 2$. We will use the estimated parameters to create a model fit, and compare to the data by plotting residuals.

Solution

(a) The estimated parameters $\theta = [\beta, \gamma, \mu] = [4.54 * 10^{-3}, 4.89 * 10^{-9}, 3.01 * 10^{-1}]$. Utilizing `fminsearch.m` produced values outside the interval $[0, 1]$. (b) We estimate the variance σ^2 to be 200.731. Using calculated sensitivities, we compute the covariance matrix to be:

$$V = \begin{bmatrix} V(\frac{\partial I}{\partial \beta}) & Cov(\frac{\partial I}{\partial \beta}, \frac{\partial I}{\partial \gamma}) & Cov(\frac{\partial I}{\partial \beta}, \frac{\partial I}{\partial \mu}) \\ Cov(\frac{\partial I}{\partial \gamma}, \frac{\partial I}{\partial \beta}) & V(\frac{\partial I}{\partial \gamma}) & Cov(\frac{\partial I}{\partial \gamma}, \frac{\partial I}{\partial \mu}) \\ Cov(\frac{\partial I}{\partial \mu}, \frac{\partial I}{\partial \beta}) & Cov(\frac{\partial I}{\partial \mu}, \frac{\partial I}{\partial \gamma}) & V(\frac{\partial I}{\partial \mu}) \end{bmatrix} = \begin{bmatrix} 0.00 & -1.00 * 10^{-7} & 1.30 * 10^{-6} \\ -1.00 * 10^{-7} & 3.20 * 10^{-6} & -4.41 * 10^{-5} \\ 1.30 * 10^{-6} & -4.41 * 10^{-5} & 6.59 * 10^{-4} \end{bmatrix}$$

We note that V is invertible and $\det(V) = 2.01 * 10^{-19} \neq 0$, so V is full-rank. However, the parameters exhibit low covariance, and as such do not appear to be very correlated. This is unexpected, as typical disease models would have correlations between transmission rate (β) and recovery rate (γ) based on the traits of the disease. However, this is not necessarily the case. (c) We compute the 95% confidence intervals: $\beta \in [0.0044, 0.0047]$, $\gamma \in [-0.0035, 0.0035]$, $\mu \in [0.25, 0.35]$. In Figure 5, we compare these confidence intervals to intervals of two standard deviations about the parameter mean. In each case, the confidence intervals closely match the $\pm 2\sigma$ intervals. (d) We now plot the distributions for each parameter (Figure 6, 7, 8). Next, we plot our model fit for $I(t_j)$ alongside the data (Figure 9). Visually, our model appears aligned with the data, but we will analyze the residuals to confirm. We plot the residuals between our best-fit model and the file data (Figure 10). As we can see, the

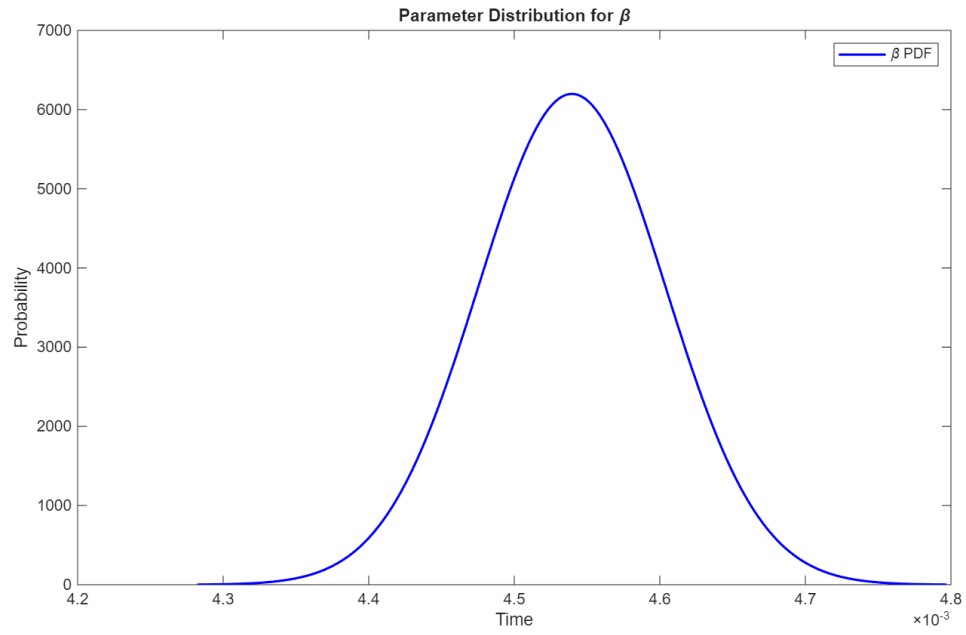


Figure 6: Parameter Distribution for β

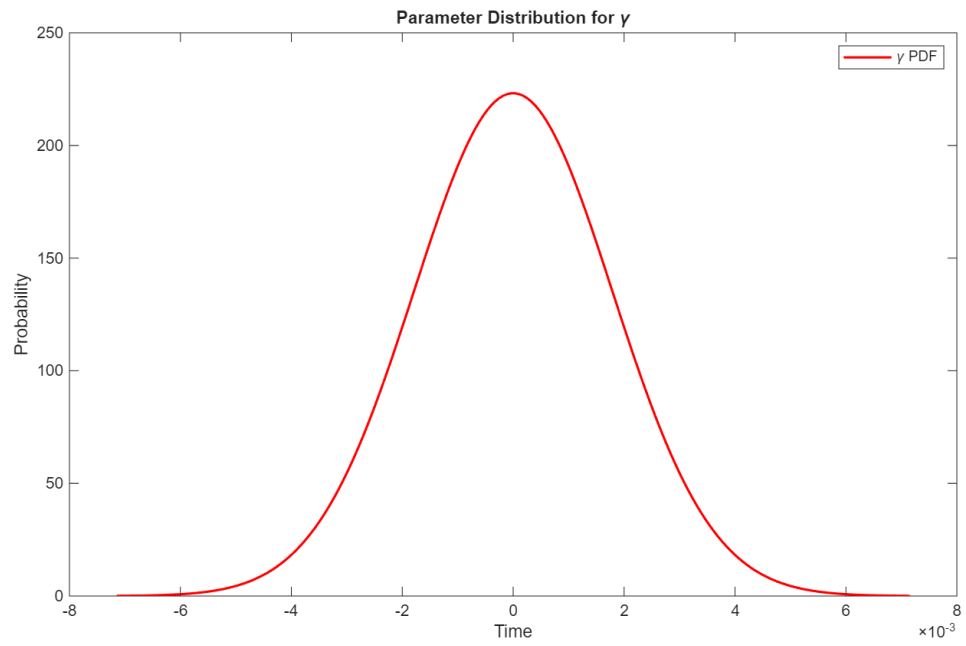


Figure 7: Parameter Distribution for γ

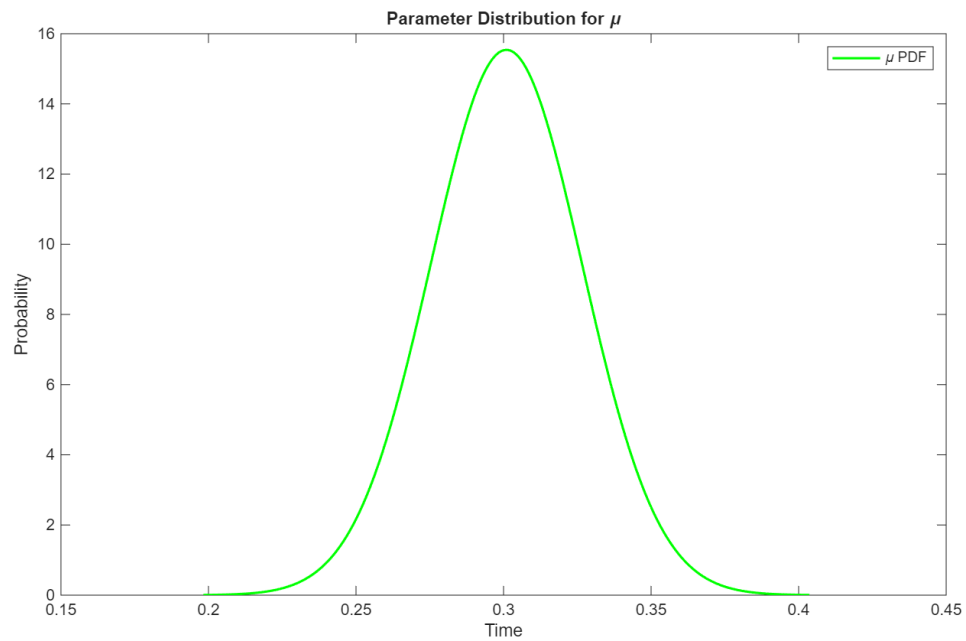


Figure 8: Parameter Distribution for μ

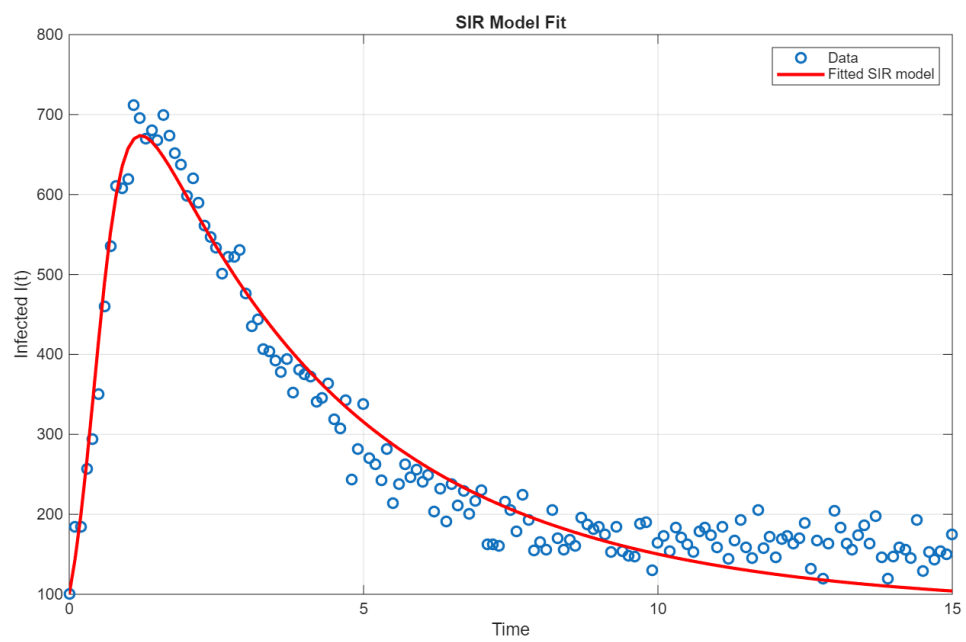


Figure 9: $I(t_j)$ Model Compared to Data

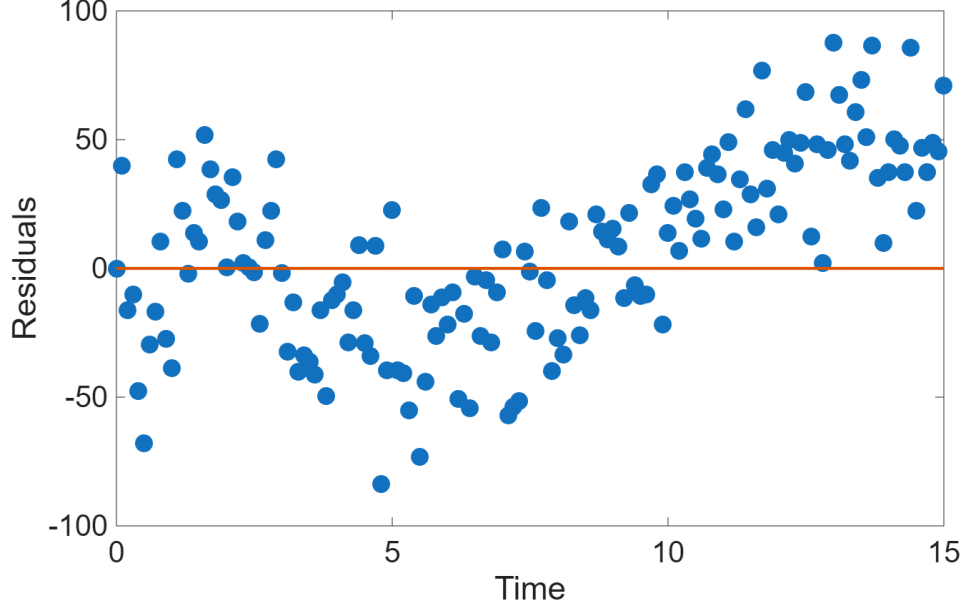


Figure 10: I Residuals

residuals do not follow any pattern, and so we can assume the error is iid and normally distributed. As such, our model is a strong fit to the data. Finally, we plot the trajectories for $S(t)$, $I(t)$, and $R(t)$, as predicted by our model (Figure 11). (e) Using the Table 1 data, we estimate the parameters to be $[\beta, \gamma] = [0.0023, 0.43]$. The variance is estimated to be $\sigma^2 = 5.61$. We compute the covariance matrix to be:

$$V = \begin{bmatrix} V(\frac{\partial I}{\partial \beta}) & Cov(\frac{\partial I}{\partial \beta}, \frac{\partial I}{\partial \gamma}) \\ Cov(\frac{\partial I}{\partial \gamma}, \frac{\partial I}{\partial \beta}) & V(\frac{\partial I}{\partial \gamma}) \end{bmatrix} = \begin{bmatrix} 0.00 & 3.00 * 10^{-9} \\ 3.00 * 10^{-9} & 3.80 * 10^{-6} \end{bmatrix}$$

Again, V is invertible and $det(V) = 1.042 * 10^{-16} \neq 0$, so V is full rank. However, the parameters appear again to not be very correlated. This is again unexpected, but the small sample size limits the applicability of this finding. We now plot our model fit against the data (Figure 12). Despite the limited dataset, our model appears to be a good fit. Now, we plot the residuals (Figure 13). Since the residuals show no pattern, we can assume the error is iid. This supports the accuracy of our model, and its ability to predict the activity of the outbreak.

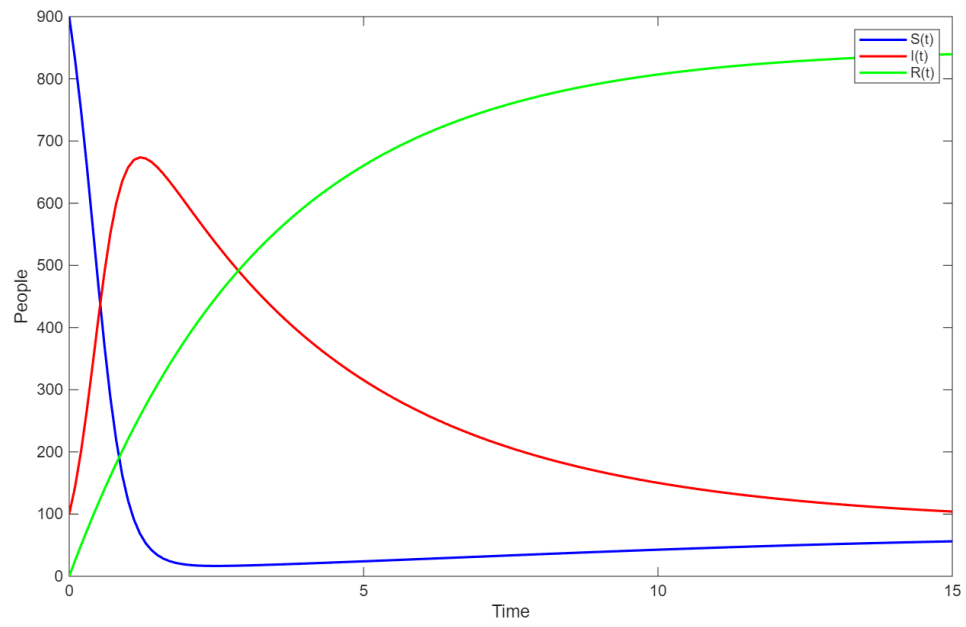


Figure 11: Trajectories for $S(t)$, $I(t)$, and $R(t)$

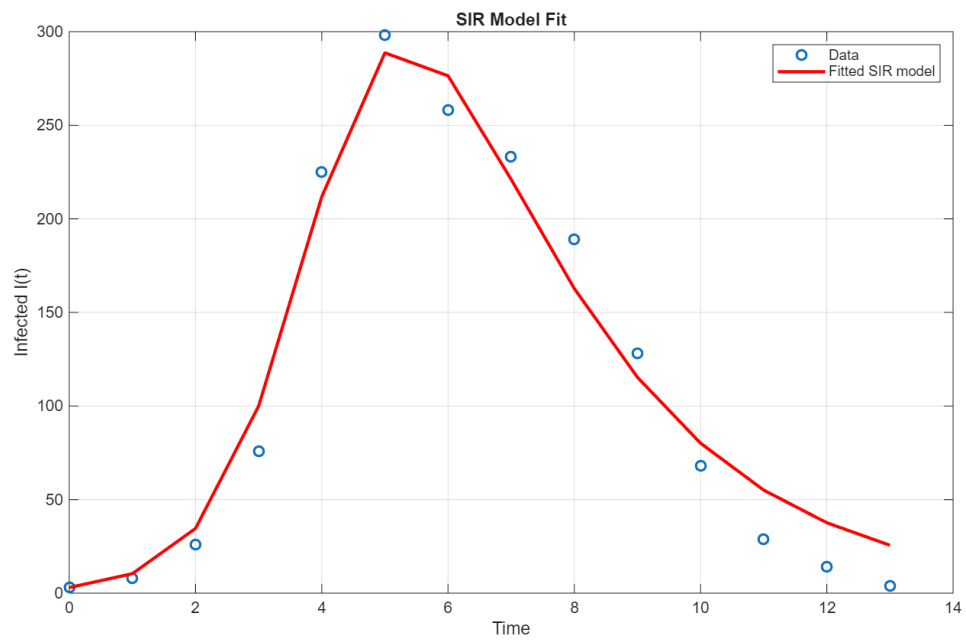


Figure 12: SIR Model Fit for Table 1

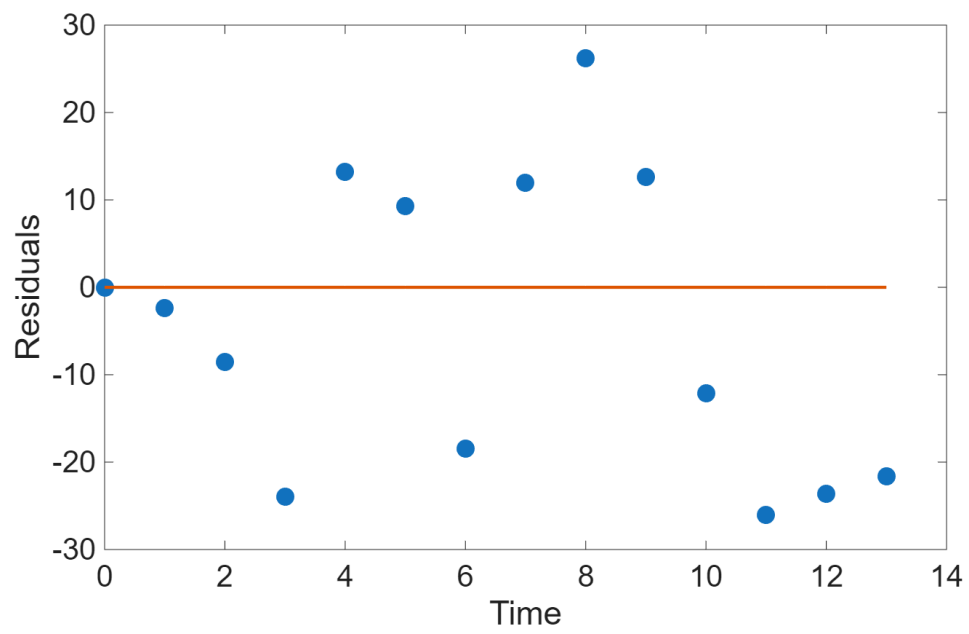


Figure 13: Residuals for Model vs Table 1